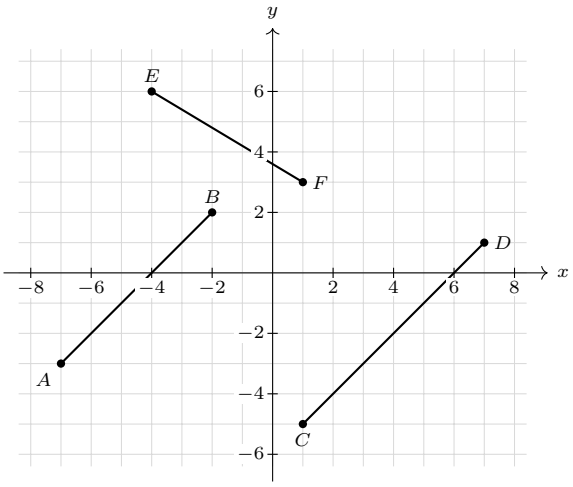
Lesson 1 — Extra Practice

Length, Distance, and Midpoint of a Line Segment

Part A — Read the Grid Before You Count It

Lesson 1 counts squares. These questions make you earn the numbers first: the endpoints have to be read off correctly, and on one grid a square is not worth 1. *Give every length as* ***(i)*** *the exact value in simplest radical form and* ***(ii)*** *a decimal to the* ***nearest tenth****. Do the exact value first, every time.*

1. Segments AB, CD and EF are graphed below. Each grid square is 1 unit.

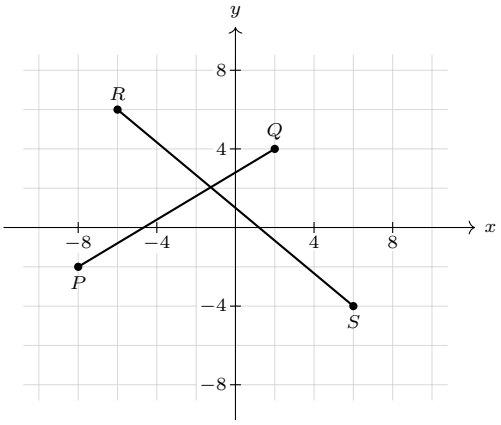


(a) State the coordinates of all six endpoints.

(b) Determine the length of AB, CD and EF.

(c) Two of these three lengths simplify and one does not. Say which, and explain what it is about the number under the radical that decides it.

2. On the grid below **each grid square is** 2 **units**, not 1. Read the tick numbers carefully before you count anything.



(a) State the coordinates of P, Q, R and S.

(b) Determine the length of PQ and the length of RS.

(c) A student counts squares as though each were worth 1 and gets $\sqrt{34}$ for PQ. Their arithmetic is perfect. Explain, using the scale, exactly how their answer is related to the correct one.

Part B — The Exact Value Comes First

3. Determine the length of each line segment. Give **(i)** the exact value in simplest radical form and **(ii)** a decimal to the nearest tenth.

(a) $A(-3,5)$ to $B(4,-2)$

(b) $C(-6,-1)$ to $D(2,3)$

(c) $E(5,-8)$ to $F(-4,4)$

(d) $G(-2,-7)$ to $H(3,-1)$

(e) $J(-8,3)$ to $K(-2,-3)$

(f) $L(4,9)$ to $M(-3,5)$

(g) Exactly one of the six answers above needed no radical sign at all in its final form. Which one, and what was special about the number under the radical?

Part C — Perimeter From Coordinates

Leave every perimeter as an exact sum of radicals first, then give the decimal to the nearest tenth. Round only at the very end — never round a side and then add.

4. $\triangle ABC$ has vertices $A(-4,-3)$, $B(2,5)$ and $C(6,-1)$.

(a) Determine the exact length of each of the three sides.

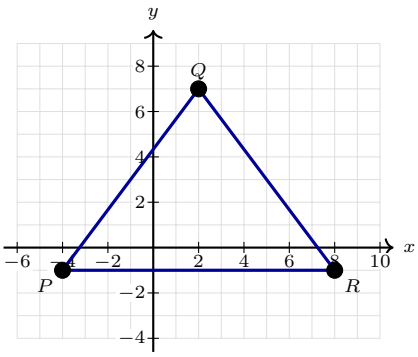
(b) State the exact perimeter as a sum of radicals, then give it to the nearest tenth.

(c) Is the triangle scalene, isosceles, or equilateral? Which tool — counting squares or the distance calculation — settles that, and why?

Part D — One Figure, Two Formulas

*The booklet usually tells you which formula to use. From here on it will not. Before you compute anything, decide what the question is actually asking for: a **length** (distance formula) or a **location** (midpoint formula). Give exact radicals in simplest form first, then a decimal to the nearest tenth where a decimal is asked for.*

5. $\triangle PQR$ has vertices $P(-4,-1)$, $Q(2,7)$, and $R(8,-1)$, as shown.



(a) Determine the perimeter of $\triangle PQR$.

(b) Determine the coordinates of the midpoint of each of the three sides.

(c) Determine the length of the segment joining the midpoint of $\overline{PQ}$ to the midpoint of $\overline{QR}$, and compare it to the length of $\overline{PR}$.

6. For each task, state which formula settles it — **distance**, **midpoint**, or **both** — and give one sentence of reasoning. Do not calculate anything.

(a) Locating the centre of a circle when the endpoints of a diameter are known.

(b) Deciding whether a triangle is isosceles.

(c) Deciding whether the diagonals of a quadrilateral bisect each other.

(d) Finding the radius of the circle in part (a).

(e) Deciding whether a given point is the same distance from two other points.

(f) Finding a fourth vertex that completes a parallelogram.

Part E — Backwards: the Midpoint Is Known

*If M is the midpoint of $\overline{AB}$, then $x_M = \dfrac{x_A + x_B}{2}$, so $x_B = 2x_M - x_A$ — **double the midpoint, then subtract the endpoint you have**. Averaging a second time is the classic wrong move; Question 12 asks you to name why.*

7. In each case M is the midpoint of $\overline{AB}$. Determine the coordinates of B.

(a) $A(-5,4)$, $M(3,-2)$ (b) $A(-9,3)$, $M(0,-6)$ (c) $A(1.5,-3)$, $M(-4.5,2)$

8. A search helicopter hovers at $H(14,-6)$, exactly halfway along the straight line from base camp $C(-3,11)$ to a stranded hiker. Determine the hiker's coordinates, then verify your answer by checking that H really is the midpoint.

Part F — Backwards: the Distance Is Known

Given a length and one endpoint, the other endpoint is usually ***not*** *unique. Squaring loses a sign, so undoing it restores* two *possibilities. Give both, every time — unless you can show there is only one, or none.*

9. Find **all** possible coordinates of the second endpoint.

(a) $A(3,-2)$, and B lies on the vertical line $x=8$, with $AB=13$.

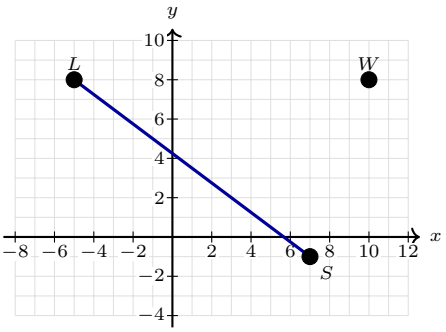
(b) $C(-4,1)$, and D lies on the horizontal line $y=6$, with $CD=\sqrt{41}$.

(c) $E(2,7)$, and F lies on the horizontal line $y=2$, with $EF=\sqrt{74}$.

(d) $G(3,-2)$, and H lies on the vertical line $x=9$, with $GH=4$. Explain your answer in terms of the shortest possible distance from G to that line.

Part G — A Map With a Scale

10. A provincial park prints a map with a coordinate grid laid over it. A fire lookout is at $L(-5,8)$, a ranger station is at $S(7,-1)$, and a warden's cabin is at $W(10,8)$. On this map 1 grid unit represents 2 km.



(a) Determine the distance from L to S in grid units, then convert it to kilometres.

(b) A supply drop is to be made at the point exactly halfway between L and S. State its coordinates.

(c) How far, in kilometres, must a crew travel from the lookout to reach the supply drop?

(d) Is the cabin W closer to L or to S? Support your answer with distances in kilometres, to the nearest tenth.

Part H — Error Analysis

11. Asked for the length of the segment from $A(-2, 5)$ to $B(4, -3)$, a student writes:

$$AB = (x_2 - x_1) + (y_2 - y_1) = \big(4 - (-2)\big) + (-3 - 5) = 6 + (-8) = -2.$$

(a) The student's subtraction is correct. Name the two steps of the distance formula they left out.

(b) Their answer is -2. Give a reason, from the meaning of the word *length*, why the answer had to be wrong before any checking.

(c) Determine the correct length of AB, exactly and to the nearest tenth.

(d) Explain what squaring does to the two differences that makes the order of the endpoints — A to B or B to A — stop mattering.

12. Asked to find B given $A(-5, 4)$ and midpoint $M(3, -2)$ — Question 7(a) — Theo averaged A and M:

$$B = \left(\frac{3 + (-5)}{2},\ \frac{-2 + 4}{2}\right) = (-1, 1).$$

(a) Explain precisely why averaging is the wrong operation here, and what Theo's point $(-1, 1)$ actually is.

(b) Test Theo's answer: is $M(3, -2)$ the midpoint of $A(-5, 4)$ and $(-1, 1)$? Show the calculation.

(c) Write, in one sentence, the rule that recovers a missing endpoint, and explain why it is the algebraic opposite of averaging.

Part I — Multiple Choice and Numerical Response

13. *(Multiple Choice)* Segment AB has endpoint $A(-5, 2)$ and length $\sqrt{89}$. Point B lies on the line $y = -3$. The possible x-coordinates of B are

A. 3 only B. -13 only C. 3 or -13 D. no such point exists

14. *(Multiple Choice)* The exact length of the segment from $(-7, -5)$ to $(5, 13)$, written as a mixed radical **in simplest form**, is

A. $2\sqrt{117}$ B. $6\sqrt{13}$ C. $3\sqrt{52}$ D. $18\sqrt{2}$

15. *(Numerical Response)* $\triangle ABC$ has vertices $A(-5,-5)$, $B(7,4)$ and $C(-5,4)$. The perimeter of $\triangle ABC$, in units, is _____. *(Record your answer in the numerical response box from left to right.)*

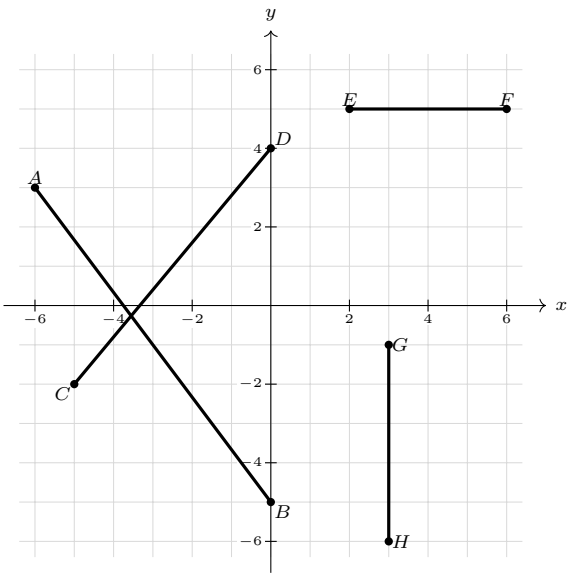
Lesson 2 — Extra Practice

Rise Over Run: The Slope of a Line Segment

Part A — Counting Slope Off a Grid

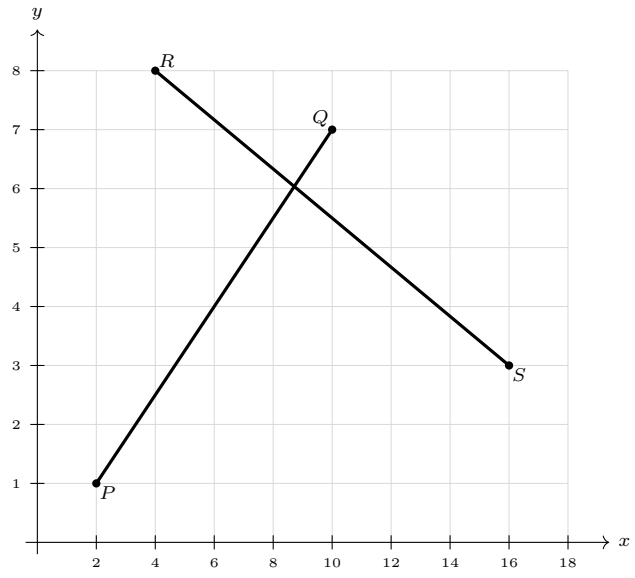
Slope is $\dfrac{rise}{run} = \dfrac{y_2 - y_1}{x_2 - x_1}$. *Counting squares only gives the slope when one square is one unit* ***on both axes*** *— question 2 is there to catch you.*

1. Each segment below has endpoints with integer coordinates. Complete the table. Write *undefined* in the slope column where that is the correct answer, and give the rise and run anyway.



Segment	Rise	Run	Slope
AB			
CD			
EF			
GH			

2. On the grid below **one square across represents 2 units**, while one square up represents 1 unit. Read the tick labels, not the squares.



(a) Write the coordinates of P, Q, R and S.

(b) Calculate the slope of PQ and the slope of RS.

(c) A student counts squares instead of units and reports $\frac{3}{2}$ for PQ. State the relationship between that answer and the correct one, and explain where the factor comes from.

(d) What answer would the same square-counting student get for RS? Would *any* segment on this grid be given its correct slope by counting squares? Explain.

3. Four segments are given by their endpoints.

$$J(-4,1) \to K(2,-5), \qquad L(-3,-2) \to M(3,7), \qquad N(-5,4) \to P(5,0), \qquad Q(1,6) \to R(3,1)$$

(a) Calculate all four slopes.

(b) List the four segments in order from **least steep to steepest**. *Steepness is measured by the size of the slope, ignoring its sign* — so compare $|m|$, not m.

(c) Which segment has the **greatest** slope as a number?

(d) Which segment has the **least** slope as a number?

(e) The segment with the greatest slope is not the steepest one. Explain, in one sentence, why "greatest slope" and "steepest" are different questions once negatives are involved.

Part B — Slope From Coordinates

4. Calculate the slope of the segment joining each pair of points. Reduce every fraction to lowest terms and carry the sign on the **numerator**.

(a) $A(-7,2)$, $B(5,-4)$

(b) $C(3,-8)$, $D(-5,-2)$

(c) $E(-6,-5)$, $F(-1,7)$

(d) $G\left(\frac{1}{2},-3\right)$, $H\left(-\frac{5}{2},4\right)$

(e) $J(-9,4)$, $K(6,4)$

(f) $L(-3,-7)$, $M(-3,5)$

(g) Three of your six answers came out negative. For each of those, say whether it was the rise or the run that was negative, and confirm that subtracting the coordinates in the *opposite* order gives the same slope.

5. For each pair of points, state whether the slope is 0 or **undefined**. Beside each answer write the value of the **denominator** $x_2 - x_1$ — that is the number that settles it.

(a) $(4, -6)$ and $(-9, -6)$

(b) $(7, 3)$ and $(7, -11)$

(c) $(0, -4)$ and $(0, 9)$

(d) $(-8, 0)$ and $(3, 0)$

(e) $(-5, 2.5)$ and $(11, 2.5)$

(f) $(-1.5, -4)$ and $(-1.5, 6)$

(g) Complete the sentence using the words *numerator* and *denominator*: "A slope of 0 happens when _____, and an undefined slope happens when _____." Then explain why $\frac{0}{7}$ is a perfectly good number but $\frac{7}{0}$ is not.

6. A line passes through the point $(5, -2)$.

(a) Give a second point so that the slope is 0.

(b) Give a second point so that the slope is undefined.

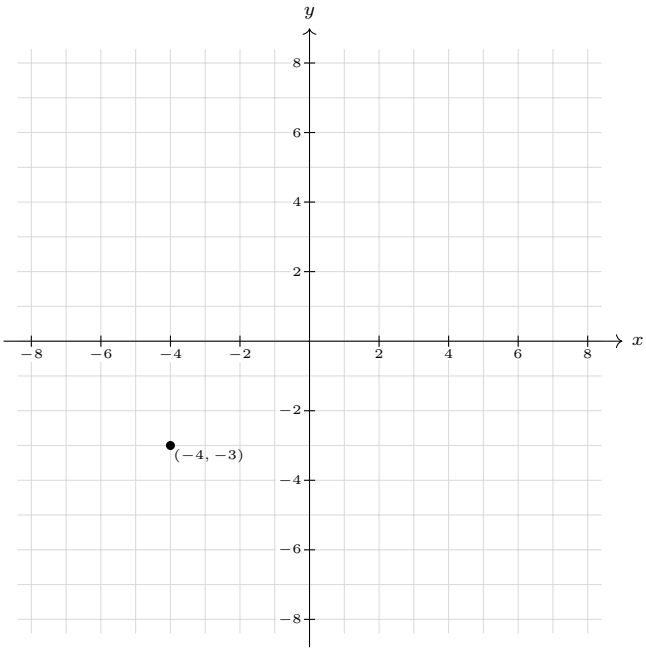
(c) Describe *every* point that would work in (a), and every point that would work in (b).

(d) Explain why no single line can have both slope 0 and an undefined slope.

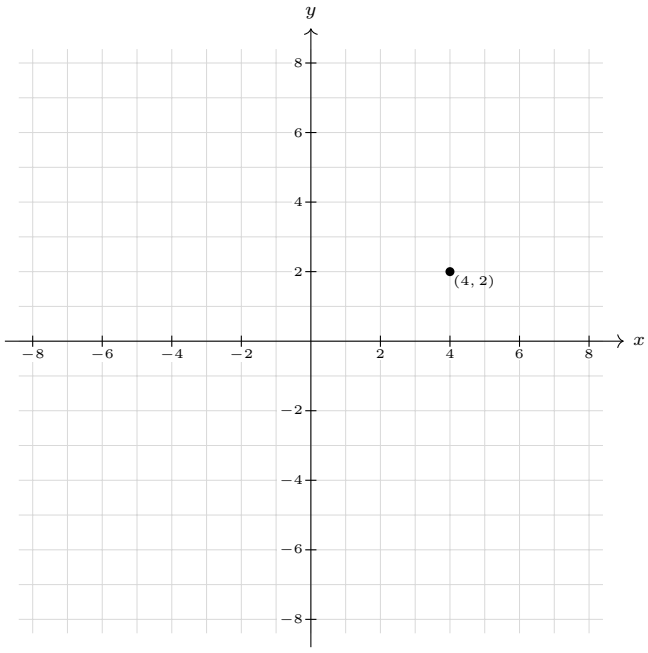
Part C — Running the Formula Backwards

7. Plot the given point, then draw the line through it with the given slope, long enough to reach both edges of the grid. Underneath, list **three** other points on your line that have integer coordinates.

(a) $(-4, -3)$, slope $\frac{5}{2}$



(b) $(4, 2)$, slope $-\frac{3}{4}$



(c) Your three points in (a) are almost certainly not the same as a classmate's, yet you can both be right. Explain what *all* the correct answers have in common, using the phrase "the same multiple of the rise and the run".

8. In each case the slope of the segment is given. Determine the missing coordinate. Check each answer by substituting back into $m = \frac{y_2 - y_1}{x_2 - x_1}$.

(a) $A(3, k)$, $B(9, -1)$, $m = -\frac{2}{3}$

(b) $P(-5, 4)$, $Q(t, -2)$, $m = \frac{3}{4}$

(c) $C(w, -7)$, $D(2, 5)$, $m = -4$

(d) $E(-6, h)$, $F(-6, 9)$, slope undefined

(e) $G(-1, y)$, $H(7, 2)$, $m = 0$

(f) Parts (d) and (e) behaved differently from (a), (b) and (c). One had exactly one answer; the other had infinitely many. Say which is which, and why the formula forced that outcome.

9. Use $m = \dfrac{\text{rise}}{\text{run}}$ in reverse.

(a) A segment starts at $(-2, 5)$, has slope $-\dfrac{3}{8}$ and a run of 24. Determine the other endpoint.

(b) A segment starts at $(7, -2)$, has slope $\dfrac{5}{6}$ and a rise of -15. Determine the other endpoint.

(c) A segment has slope $\dfrac{4}{9}$, and both its rise and its run are negative. In which direction does the second endpoint lie from the first? Justify from the signs.

Part D — Slope as a Rate, With Units

10. A loading ramp rises 0.9 m over a horizontal run of 7.2 m.

(a) Determine the slope of the ramp as a fraction in lowest terms.

(b) Engineers call this the *grade* and write it as a percent. Express the slope as a percent grade.

(c) The slope is a rate. State the rate in words, with units, in the form "____ metres up per ____ metres along".

(d) A second ramp has the same grade but a run of 10 m. How high does it rise?

11. A wheelchair ramp must have a slope of **no more than** $\dfrac{1}{12}$ — at most 1 m of rise for every 12 m of run. A doorway threshold sits 0.75 m above the sidewalk.

(a) Determine the **shortest** horizontal run a single straight ramp to the threshold may have.

(b) A contractor has only 7.5 m of sidewalk to work with. Calculate the slope that run would produce, and decide whether it meets the rule. Justify by comparing the two slopes.

(c) If the contractor must use that 7.5 m run, what is the greatest rise the ramp may have?

(d) The rule is a *maximum* slope, yet it forces a *minimum* length. Explain why a larger slope is the dangerous direction here, and rearrange $\dfrac{\text{rise}}{\text{run}} \le \dfrac{1}{12}$ into a statement about the run.

12. A roof described as a "4 in 12" pitch rises 4 ft vertically for every 12 ft of horizontal run.

(a) State the slope of that roofline in lowest terms.

(b) The far side of the roof is the mirror image. State its slope and explain the sign.

(c) The house is 30 ft wide and the peak is centred. How far above the eaves is the peak?

(d) A "9 in 12" roof is quoted for a different house. Which roof is steeper, and by what factor?

Part E — Error Analysis

13. Asked for the slope of the segment joining $A(-3, 5)$ and $B(5, -1)$, a student writes

$$m = \frac{x_2 - x_1}{y_2 - y_1} = \frac{5 - (-3)}{-1 - 5} = \frac{8}{-6} = -\frac{4}{3}.$$

(a) Name the mistake in one sentence.

(b) Determine the correct slope.

(c) State the relationship between $-\frac{4}{3}$ and your answer.

(d) For which slopes would this mistake go unnoticed — that is, when does run over rise equal rise over run?

14. Asked for the slope of the segment joining $P(-2, 7)$ and $Q(6, 1)$, a student computes the rise as 6 and the run as 8 and answers $\frac{3}{4}$.

(a) The run is right. Show the correct rise, with its sign, and give the correct slope.

(b) Describe the line the student's answer would produce, and the line the correct answer produces, in terms of climbing or falling from left to right.

(c) Without any calculation, how could a quick sketch of P and Q have caught this error?

15. Asked for the slope of the segment joining $(4, -3)$ and $(4, 9)$, a student writes: "The run is 0. Zero divided into anything is zero, so the slope is 0."

(a) Write the rise and the run, then the fraction the formula actually produces.

(b) The fraction is not 0. Explain, in terms of the **denominator**, what it is instead.

(c) Give two points whose segment really does have slope 0, and say how that segment differs from this one.

(d) Why is "no slope" also an unacceptable answer for this segment?

Part F — Stretch

16. Determine the value of k in each case, or state that every value of k works.

(a) The segment joining $A(2, k)$ and $B(-4, 3)$ has slope $-\frac{1}{2}$.

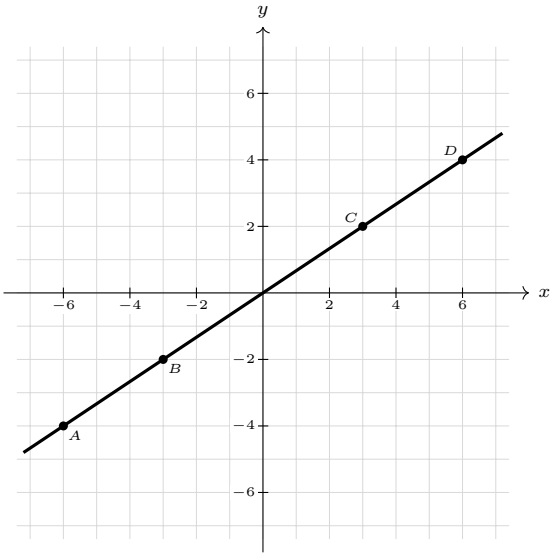
(b) The segment joining $A(k, -1)$ and $B(k+6, 8)$ has slope $\frac{3}{2}$.

(c) The segment joining $A(k, 2k)$ and $B(6, -3)$ has slope -1.

(d) The segment joining $A(k, 1)$ and $B(3, k)$ has slope $-\frac{1}{2}$.

(e) Part (b) is different from the others. Explain what is special about those two points, and what it tells you about how the slope depends on k.

17. The points $A(-6, -4)$, $B(-3, -2)$, $C(3, 2)$ and $D(6, 4)$ all lie on one line.



(a) Calculate the slope using A and B, then using A and D, then using B and C.

(b) On the grid, draw the rise-and-run right triangle for AB and the rise-and-run right triangle for AD. Make each triangle's legs run parallel to the axes.

(c) Explain why those two triangles must be **similar**. Name the three angles that match, and say why the hypotenuses have the same direction.

(d) Similar triangles have proportional sides. Use that fact to argue that *any* two points on this line give the same slope — so slope is a property of the line, not of the pair of points chosen.

18. *(Multiple Choice)* A line segment has slope $-\frac{3}{5}$. Which statement must be true?

A. A run of 5 along the segment goes with a rise of 3.

B. A run of -10 along the segment goes with a rise of 6.

C. The segment is less steep than a segment of slope $\frac{1}{2}$.

D. The segment rises from left to right.

19. *(Numerical Response)* The segment joining $(-4, p)$ and $(8, 20)$ has a slope of $\frac{3}{2}$. The value of p is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 3 — Extra Practice

Slope Relationships: Collinearity, Parallel Lines, and Perpendicular Lines

Part A — Three Ways to Test Collinearity

Three points are collinear when they lie on one straight line. Slope settles it: if two of the three slopes agree, all three points are on one line. Distance settles it too — the two short pieces must add to the long one — but only if you know which point is in the middle. Every question in this part asks you to say ***which tool*** *you used and* ***what it proved****.*

1. Consider the points $A(-8,-5)$, $B(-2,-1)$ and $C(7,5)$.

(a) Test the three points for collinearity by comparing m_{AB} with m_{BC}.

(b) Test them again by comparing m_{AB} with m_{AC}.

(c) Test them a third way: find the exact lengths AB, BC and AC, and check whether $AB + BC = AC$.

(d) All three tests reach the same verdict. Which one was the fastest to carry out, and what exactly made it faster than the other two?

(e) Suppose the three points had *not* been collinear. Describe what each of the three tests would have shown instead.

2. Fatima is testing $D(-4,7)$, $E(8,-5)$ and $F(2,1)$ for collinearity with the distance formula. She calculates $DE = 12\sqrt{2}$, $EF = 6\sqrt{2}$ and $DF = 6\sqrt{2}$, then writes:

"$DE + EF = 12\sqrt{2} + 6\sqrt{2} = 18\sqrt{2}$, but $DF = 6\sqrt{2}$. Since $18\sqrt{2} \neq 6\sqrt{2}$, the three points are not collinear."

(a) Her three lengths are all correct. Use slopes to decide whether her conclusion is correct.

(b) Which of the three points actually lies between the other two? Write the equation that the distance test should have checked, and check it.

(c) State, in one sentence, what the distance test requires that the slope test does not.

3. Decide whether each set of three points is collinear. Name the tool you used and show the comparison that settles it.

(a) $A(-4,-9)$, $B(0,-3)$, $C(6,6)$

(b) $P(5,-2)$, $Q(5,7)$, $R(5,13)$

(c) $D(-3,4)$, $E(2,-1)$, $F(8,-6)$

(d) $G(-7,3)$, $H(1,3)$, $I(15,3)$

(e) In parts (b) and (d) you never wrote a fraction such as $\frac{3}{4}$. Explain how you justified collinearity in each of those two cases, and why the argument is still a slope argument.

Part B — Solving for k

In the booklet you were handed two points and asked for a slope. Here the slope relationship is handed to you and one coordinate is missing. Set up the equation the relationship demands, then solve it.

4. Determine the value of k that makes the three points **collinear**.

(a) $A(-3, 2)$, $B(1, k)$, $C(9, 14)$

(b) $P(k, 5)$, $Q(2, -3)$, $R(6, -11)$

(c) $A(-2, -8)$, $B(4, 10)$ and the point $M(k, k)$

(d) $O(0, 0)$, $S(3, k)$, $T(k, 12)$
(two answers — find both)

5. Determine the value of k that makes $AB \parallel CD$.

(a) $A(-4, 1)$, $B(2, 5)$, $C(0, -3)$, $D(9, k)$

(b) $A(k, -2)$, $B(4, 7)$, $C(-6, 3)$, $D(-2, -9)$

(c) $A(-5, k)$, $B(3, 2k)$, $C(1, -4)$, $D(7, 5)$

(d) $A(6, 1)$, $B(6, -4)$, $C(k, 4)$, $D(2k + 3, 10)$
(Careful — m_{AB} is not a number.)

6. Determine the value of k that makes $AB \perp CD$.

(a) $A(-2, -1)$, $B(4, 3)$, $C(1, 8)$, $D(k, 2)$

(b) $A(-7, k)$, $B(5, 1)$, $C(0, -2)$, $D(8, 4)$

(c) $A(3, -4)$, $B(k, 2)$, $C(-1, 5)$, $D(5, -7)$

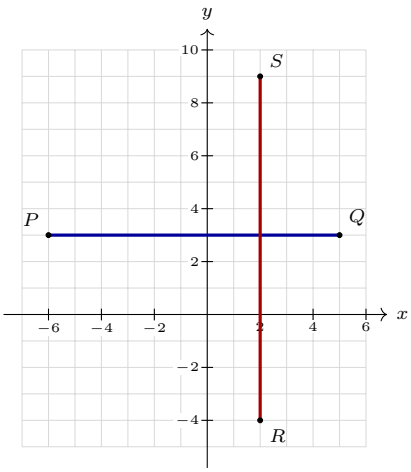
(d) $A(-4, k)$, $B(9, k)$, $C(2, -5)$, $D(2, 7)$

(e) Part (d) does not behave like the others. Say what its answer is, and explain what it is about AB and CD that produces that answer.

Part C — The One Case the Product Rule Cannot Handle

7. Segment PQ joins $P(-6, 3)$ to $Q(5, 3)$. Segment RS joins $R(2, -4)$ to $S(2, 9)$. Adeyemi looks at the figure and writes:

"$m_{PQ} = 0$ and m_{RS} is undefined, so $m_{PQ} \times m_{RS}$ is not -1. Perpendicular lines must have slopes with product -1. Therefore PQ and RS are not perpendicular."



(a) Calculate m_{PQ} and m_{RS}. Adeyemi's two slopes are correct — confirm them.

(b) Look at the figure. At what angle do PQ and RS actually cross? Are they perpendicular?

(c) Adeyemi's arithmetic is not the problem. Identify the step in the argument that fails, and say why $m_{PQ} \times m_{RS}$ cannot be evaluated at all.

(d) Rewrite the perpendicular rule as a two-part statement, so that a student who follows it would classify PQ and RS correctly. Your statement must cover both the ordinary case and the exception.

8. Each row gives the slopes of two segments. Classify the pair as **parallel**, **perpendicular**, or **neither**, and state in a few words how you decided.

(a) $m_1 = 0$, $m_2 = 0$ (b) $m_1 = 0$, m_2 undefined (c) m_1 undefined, m_2 undefined

(d) $m_1 = 0$, $m_2 = 5$ (e) m_1 undefined, $m_2 = -\dfrac{1}{4}$ (f) $m_1 = -\dfrac{7}{2}$, $m_2 = -\dfrac{2}{7}$

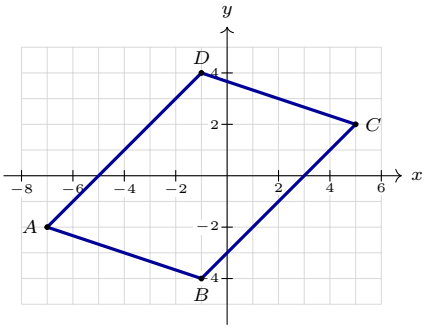
(g) Exactly one row above is a pair of perpendicular segments whose slopes do *not* multiply to -1. Name it, and explain why that row is not a counterexample to the rule.

Part D — Naming a Quadrilateral From Its Vertices

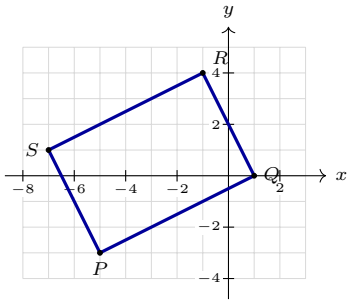
Slope proves that sides are parallel and that angles are right. Distance proves that sides are equal. Neither tool does the other's job, so a full classification needs both.

9. Four quadrilaterals are graphed below. For each one, calculate what you need and then name it as precisely as you can: **parallelogram**, **rectangle**, **rhombus**, **square**, or **trapezoid**. **Beside every claim you make, name the tool — slope or distance — that proves it.**

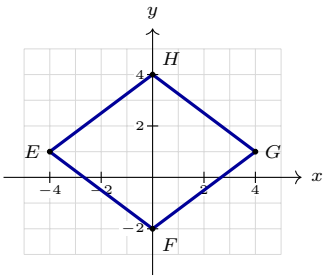
(a) $A(-7,-2)$, $B(-1,-4)$, $C(5,2)$, $D(-1,4)$



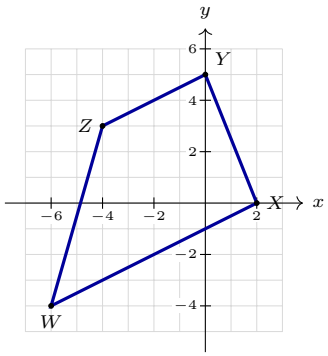
(b) $P(-5,-3)$, $Q(1,0)$, $R(-1,4)$, $S(-7,1)$



(c) $E(-4,1)$, $F(0,-2)$, $G(4,1)$, $H(0,4)$



(d) $W(-6,-4)$, $X(2,0)$, $Y(0,5)$, $Z(-4,3)$



10. Answer each question about the *evidence* a classification needs.

(a) A student shows two pairs of opposite sides are parallel and stops, calling the shape a rectangle. What is missing, and which tool supplies it?

(b) A student shows all four sides are equal and calls the shape a square. What is missing, and which tool supplies it?

(c) Why is "exactly one pair of parallel sides" a claim that needs **four** slopes, not two?

Part E — Right Triangles: Prove It, Then Measure It

Once slopes have proved that two sides meet at a right angle, those two sides are the base and the height — and $A = \frac{1}{2}bh$ works with no extra effort.

11. Consider $\triangle ABC$ with vertices $A(2, 1)$, $B(-2, 3)$ and $C(1, 9)$.

(a) Calculate all three slopes and use them to prove the triangle is right-angled. State at which vertex.

(b) Find the exact lengths of the two sides that meet at the right angle.

(c) Calculate the area of $\triangle ABC$. Explain why you did not need the third side.

12. Consider $\triangle PQR$ with vertices $P(1, -4)$, $Q(3, 2)$ and $R(-8, -1)$.

(a) Prove by slope that $\triangle PQR$ is right-angled, and state at which vertex.

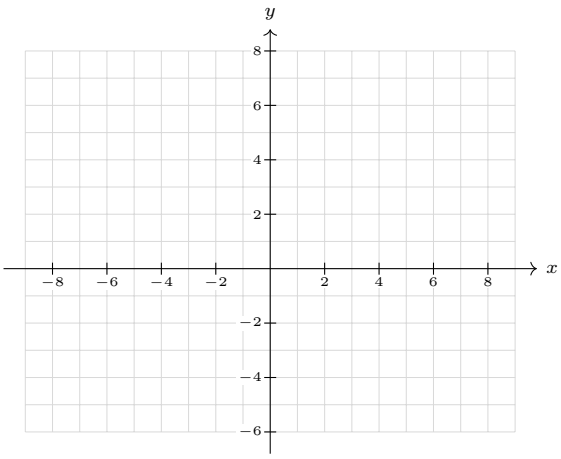
(b) Calculate the area of $\triangle PQR$.

(c) Confirm the right angle a second way, using the three side lengths and the Pythagorean theorem.

Part F — The Odd Point Out

13. Four of the five points below lie on one straight line. One does not.

$A(-8, -5)$ $B(-4, -2)$ $C(0, 1)$ $D(4, 5)$ $E(8, 7)$



(a) Plot all five points on the grid.

(b) Identify the point that is **not** collinear with the rest. Support your answer with slope calculations — a sketch on its own is not a proof.

(c) Keeping its x-coordinate, what should that point's y-coordinate have been for all five points to be collinear?

(d) A classmate tests the set by computing only m_{AB} and m_{DE}, gets $\frac{3}{4}$ and $\frac{1}{2}$, and announces that A or B must be the odd point out. Explain why that pair of slopes cannot identify the culprit, and state the smallest number of slope calculations that will.

14. The points $J(-3, 7)$, $K(-3, -2)$, $L(-3, 0)$ and $N(-2, 5)$ are given.

(a) Which point is not collinear with the other three?

(b) Every slope you can form from the other three points is *undefined*. Explain how "undefined = undefined" still proves those three are collinear here.

(c) Explain why "undefined = undefined" would **not** prove collinearity if the four points did not all share an x-coordinate.

Part G — Error Analysis

15. Asked for the slope of a segment perpendicular to a segment of slope $\frac{3}{7}$, a student answers $\frac{7}{3}$.

(a) Half of the instruction was carried out correctly. Which half?

(b) Multiply $\frac{3}{7}$ by the student's answer. What does the product tell you about the two segments?

(c) State the correct perpendicular slope and verify it with the product rule.

(d) The words "negative reciprocal" contain both instructions. Write a one-sentence check the student could run on any answer — a check that would have caught this one immediately.

16. Asked whether AB and CD are parallel, where $A(-5, 2)$, $B(1, 6)$, $C(3, -1)$ and $D(7, -7)$, a student calculates $AB = 2\sqrt{13}$ and $CD = 2\sqrt{13}$ and concludes: *"The segments are the same length, so they are parallel."*

(a) Both lengths are correct. Calculate m_{AB} and m_{CD} and state the true relationship between the two segments.

(b) Explain why equal length can never establish that two segments are parallel. Sketch two equal segments that are not parallel if it helps.

(c) Complete the two sentences: "Distance can prove _____." and "Slope can prove _____."

Part H — Stretch, Multiple Choice, and Numerical Response

17. *(Stretch)* A is the point $(-6, 2)$ and B is the point $(2, 6)$. A third point C is to be chosen so that $\angle ABC$ is a right angle.

(a) The right angle is at B. Which two segments must be perpendicular, and what slope must the second one have?

(b) Determine the coordinates of C if C lies on the x-axis. Verify your answer with the product rule.

(c) Determine the coordinates of C if instead C lies on the y-axis.

(d) There are infinitely many points C that make $\angle ABC$ a right angle, yet only one of them sits on the x-axis. Explain why, and describe when an axis would contain *no* such point.

18. *(Multiple Choice)* Segment AB joins $A(-3, 4)$ and $B(5, -2)$. Which segment below is perpendicular to AB?

A. $(1, 1)$ to $(4, 5)$ B. $(0, 0)$ to $(4, -3)$ C. $(2, 2)$ to $(5, -2)$ D. $(-1, 0)$ to $(3, 3)$

19. *(Numerical Response)* The segment joining $A(-4, 7)$ and $B(k, -5)$ is perpendicular to the segment joining $C(2, 1)$ and $D(8, 5)$. The value of k is _____. *(Record your answer, digit by digit, in the boxes below.)*

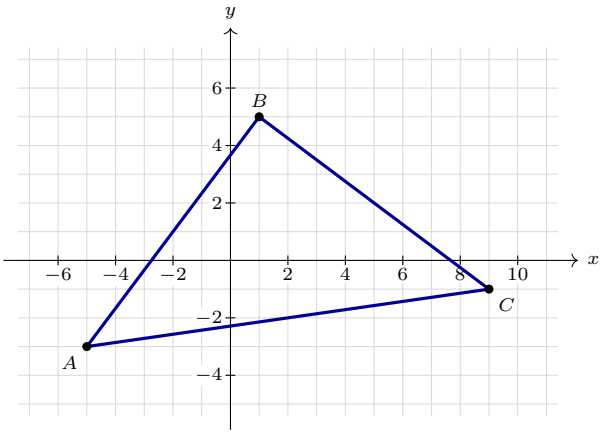
Lesson 5 — Extra Practice

Characteristics of Linear Relations Unit Review

Part A — One Figure, Every Tool

This set is the unit's review, so every question below needs ***at least two*** *of the three tools. Keep the conventions you have been using all unit: a length is an exact simplified radical first and then a decimal to the nearest tenth; a midpoint is an ordered pair; a slope is reduced to lowest terms with the sign on the numerator, and a vertical segment's slope is* undefined. *Before each calculation, say which tool you are reaching for and what claim it settles.*

1. $\triangle ABC$ has vertices $A(-5,-3)$, $B(1,5)$ and $C(9,-1)$, graphed below. This one triangle is taken through everything the unit teaches.



(a) Determine the exact length of AB, BC and AC, and give each to the nearest tenth.

(b) State the exact perimeter as a sum, then round it once, at the end, to the nearest tenth.

(c) Determine the midpoint of each of the three sides.

(d) Classify $\triangle ABC$ as scalene, isosceles or equilateral. Name the tool that settles this, and say why the other two tools cannot.

(e) Show **two different ways** that $\triangle ABC$ has a right angle, and name the vertex. One way must use slopes; the other must use the converse of the Pythagorean theorem.

(f) Determine the area of $\triangle ABC$. Explain why your answers to (a) and (e) together make this a one-line calculation.

(g) A **median** joins a vertex to the midpoint of the opposite side. Determine the exact length *and* the slope of the median from B.

(h) Compare the slope of the median from B with the slope of AC. What does that comparison prove about the median, and how does it give the area of (f) a *second* time?

(i) Join the midpoint of AB to the midpoint of BC. Determine that segment's length and slope, and state its two relationships to side AC.

Part B — Choosing the Right Tool

2. For each claim, state which tool — **distance**, **midpoint**, **slope**, or a named combination — settles it, and give one sentence of reasoning. **Calculate nothing.**

(a) A quadrilateral has exactly one pair of parallel sides.

(b) A triangle contains a right angle, without finding any side length.

(c) A parallelogram is in fact a rhombus.

(d) One segment bisects another.

(e) A point lies on the perpendicular bisector of a segment. (Two different tools each settle this one — give both.)

(f) Four points, joined in order, enclose a figure with no crossing sides — but one of them is not where it should be, and you must say *how far* out it is.

3. For each pair of segments, answer all three questions: are they **parallel, perpendicular, or neither**; are they **equal in length**; do they **share a midpoint**?

(a) AB: $A(-6,1)$, $B(2,7)$
CD: $C(-1,-4)$, $D(7,2)$

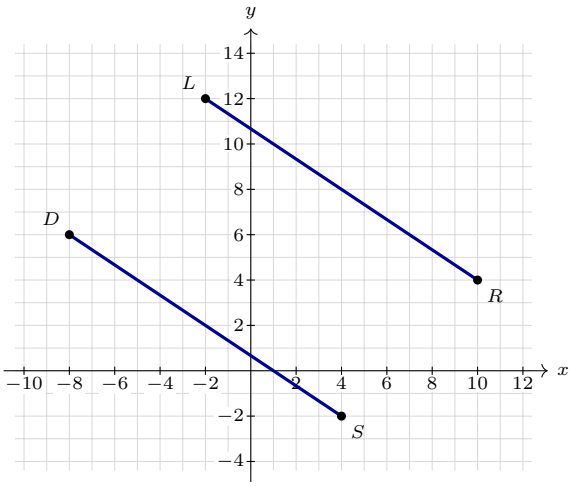
(b) EF: $E(-3,5)$, $F(5,-1)$
GH: $G(-2,-2)$, $H(4,6)$

(c) JK: $J(-4,-6)$, $K(-4,5)$
LM: $L(-9,3)$, $M(2,3)$

(d) In part (b) the four endpoints E, G, F, H, joined in that order, form a well-known shape. Name it, and say which **three** of your answers in (b) prove it.

(e) Part (c) is the one case where the product of the slopes cannot be used. State the two slopes, explain why their product is not -1 — or any number — and say how you know the segments are perpendicular anyway.

4. A city map is printed on a coordinate grid on which 1 **unit represents** 250 **m**. The bus depot is at $D(-8, 6)$, the school is at $S(4, -2)$, the library is at $L(-2, 12)$, and the rink is at $R(10, 4)$.



(a) Determine the straight-line distance from the depot to the school, first in grid units (exact, then to the nearest tenth) and then in **kilometres** to the nearest tenth.

(b) A crossing guard is posted at the point exactly midway along DS. State its coordinates, and state how far, in kilometres, it is from the depot.

(c) The bus runs $D \to S$; a second bus runs $L \to R$. Are the two routes parallel? Settle it with slopes, then show the two routes are *not* the same road by testing whether L lies on route DS.

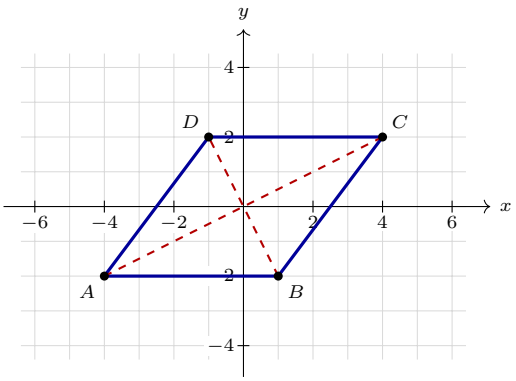
(d) Compare the lengths of the two routes. What single word describes quadrilateral $DSRL$, and which two facts have you just established that prove it?

(e) Confirm your answer to (d) a second way, using only the diagonals DR and SL. Then use their lengths to say what $DSRL$ is **not**.

Part C — Proving a Shape Is What It Claims

A name is a claim, and every claim needs the tool that settles it: slope for parallel sides and right angles, distance for equal sides, midpoint for bisecting diagonals. A shape's most complete name is the one you have evidence for — not the one the picture suggests.

5. Quadrilateral $ABCD$ has vertices $A(-4,-2)$, $B(1,-2)$, $C(4,2)$ and $D(-1,2)$. Its diagonals are drawn dashed.



(a) Determine the slope of all four sides. Which pairs are parallel?

(b) Determine the length of all four sides. What does that add to (a)?

(c) Test one pair of **adjacent** sides for perpendicularity. What does the result rule out?

(d) Determine the midpoint and the exact length of each diagonal. State what each of those two results proves.

(e) Determine whether the diagonals are perpendicular, and give $ABCD$ its most complete name. Every one of (a)–(d) must appear in your justification.

6. Quadrilateral $PQRS$ has vertices $P(-5,-2)$, $Q(5,-2)$, $R(3,4)$ and $S(-3,4)$.

(a) Determine the slope of all four sides and show that exactly **one** pair is parallel. What does that make $PQRS$?

(b) Determine the length of all four sides. Which two are equal, and what word does that add to the name?

(c) Determine the exact length of both diagonals PR and QS.

(d) A classmate sees your answer to (c) and says "equal diagonals, so $PQRS$ is a rectangle." Disprove it with **one** midpoint comparison, and state the extra condition equal diagonals need before "rectangle" is earned.

(e) Is any angle of $PQRS$ a right angle? Settle it with slopes, and say which pair of sides you tested and why that pair was enough.

7. $\triangle GHJ$ has vertices $G(-6, 0)$, $H(0, 8)$ and $J(4, 5)$.

(a) Determine the exact length of each side and classify the triangle by its sides.

(b) Prove with slopes that there is a right angle, and name the vertex. Confirm with the converse of the Pythagorean theorem.

(c) Determine the area of $\triangle GHJ$.

(d) Determine the midpoint M of the hypotenuse, then calculate MG, MH and MJ exactly. State what you notice, and how it relates to the length of the hypotenuse.

(e) A circle drawn through all three vertices of $\triangle GHJ$ has M as its centre. Use your answers in (d) to justify that, and state the circle's radius to the nearest tenth.

Part D — Working Backwards

8. Three vertices of parallelogram $ABCD$, named **in order**, are $A(-5, -2)$, $B(3, -4)$ and $C(7, 4)$.

(a) Use the fact that the diagonals of a parallelogram bisect each other to determine D. Show the midpoint calculation that produced it.

(b) Verify your D two further ways: show AB and DC have equal slopes, and show they have equal lengths. Do the same for BC and AD.

(c) Determine the exact length of each diagonal. Is $ABCD$ a rectangle? Justify with the diagonals *and* with a slope test.

(d) Three vertices of a parallelogram that are *not* named in order can give **three** possible fourth vertices. Here there is only one. What wording in this question removed the other two?

9. A is the point $(-2, -1)$, B is the point $(6, 3)$, and C is the point $(0, k)$.

(a) Determine the value of k for which $\triangle ABC$ has a right angle **at** A. Check your answer with the converse of the Pythagorean theorem.

(b) Determine **all** values of k for which $\triangle ABC$ has a right angle **at** C. Check each one.

(c) Part (a) had one answer and part (b) had two. Explain what it is about "the right angle is at C" that produces a quadratic.

(d) Determine the value of k for which A, B and C are **collinear** — that is, for which there is no triangle at all. Which tool settles this, and why is it the same tool as in part (a)?

10. A is the point $(-6, 2)$ and B is the point $(4, 8)$.

(a) One point equidistant from A and B is obvious. Name it, and say which tool produced it.

(b) Determine the slope of AB, then the slope of any line perpendicular to AB.

(c) Starting from your point in (a), step off one run and one rise of that perpendicular slope to reach a second point. State its coordinates.

(d) Verify with the distance formula that your point in (c) really is equidistant from A and B. Give both distances exactly.

(e) Describe *all* the points equidistant from A and B, and state the two pieces of information — one from a midpoint, one from a slope — that pin that set down.

11. $M(-2, 3)$ is the midpoint of $\overline{ST}$, and S is the point $(4, -5)$.

(a) Determine the coordinates of T.

(b) Check your answer with distances: calculate SM, MT and ST exactly and say what the three numbers confirm.

(c) Equal distances alone would not prove M is the midpoint. State the second condition, and the tool that checks it.

Part E — Reasoning and Error Analysis

12. A student writes: "AB joins $A(-4, -3)$ to $B(2, 1)$ and CD joins $C(-1, -3)$ to $D(5, 1)$. Both slopes are $\frac{2}{3}$, so AB and CD are the same line."

(a) Confirm that the two slopes really are equal.

(b) The conclusion is still false. Test whether C lies on line AB by comparing m_{AC} with m_{AB}, and report what you find.

(c) Rewrite the student's sentence so that it is true, using the words *parallel* and *distinct*.

(d) State the one extra check that upgrades "equal slopes" to "same line," and explain why one shared point is enough to settle it.

13. Decide whether each statement is true or false. If it is false, give a **counterexample with coordinates** and the calculation that kills it.

(a) "A rhombus is always a square." *(Question 5 supplies the counterexample if it is false.)*

(b) "A square is always a rhombus."

(c) "If the diagonals of a quadrilateral are perpendicular, it is a rhombus." *(Try $K(0, 6)$, $L(3, 2)$, $M(0, -4)$, $N(-3, 2)$.)*

(d) "If two segments have equal lengths and share a midpoint, their four endpoints form a rectangle." *(Question 3(b) is evidence one way; can you see why it is not a square in general?)*

14. A student claims: "The midpoint of the hypotenuse is the centre of the triangle." Test the claim on $\triangle GHJ$ from question 7, with $G(-6, 0)$, $H(0, 8)$, $J(4, 5)$.

(a) The claim is *half* right. Using question 7(d), state precisely which "centre" the midpoint of the hypotenuse really is.

(b) The balance point of a triangle — its **centroid** — is the average of all three vertices, $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$. Calculate it for $\triangle GHJ$.

(c) Show that the centroid is not the midpoint of the hypotenuse, and check that it is not equidistant from the three vertices by comparing just two distances.

(d) Rewrite the student's claim as two separate true statements, each naming its own centre. Why is the bare word "centre" unsafe in geometry?

15. Asked whether $PQRS$ with $P(-5, -2)$, $Q(5, -2)$, $R(3, 4)$, $S(-3, 4)$ is a parallelogram, a student writes:

$$m_{PQ} = 0 = m_{SR} \;\Rightarrow\; PQ \parallel SR; \qquad QR = PS = 2\sqrt{10} \;\Rightarrow\; QR \parallel PS; \qquad \text{so } PQRS \text{ is a parallelogram.}$$

(a) Both computed facts are correct. Identify the step that does not follow, in one sentence.

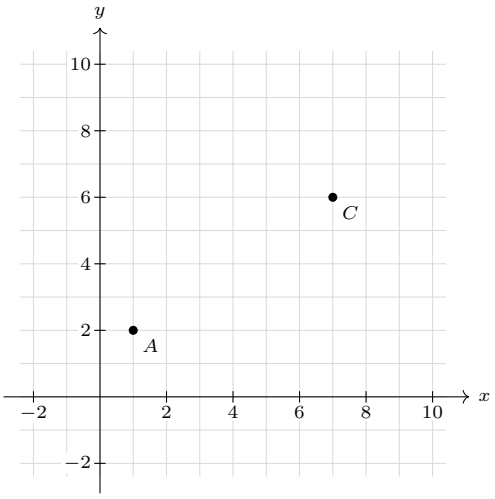
(b) Give the two calculations from question 6 that show the conclusion is wrong.

(c) State the correct pair of conditions for a parallelogram, using *both* pairs of opposite sides.

(d) Equal lengths do not force parallel, and parallel does not force equal length. Give a two-point example of each failure, with coordinates.

Part F — Challenge

16. $A(1, 2)$ and $C(7, 6)$ are the **opposite** vertices of a square $ABCD$.



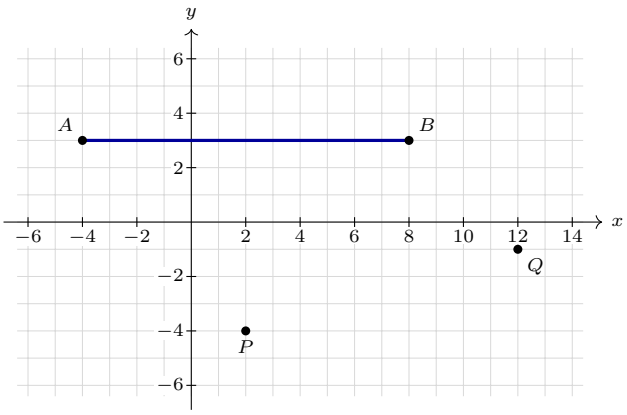
(a) Determine the centre of the square, and the exact length of diagonal AC.

(b) The diagonals of a square are equal, bisect each other, and are perpendicular. Use all three facts to locate B and D, and plot the square on the grid.

(c) Verify your square: show all four sides are equal and one angle is right.

(d) Determine the area of the square two ways — from the side length, and from $\frac{1}{2}d_1d_2$ — and confirm they agree.

17. Segment AB joins $A(-4, 3)$ and $B(8, 3)$. A drone must fly from a point to the *nearest point of the segment* — not the nearest endpoint, unless that happens to be the same thing.



(a) State the slope of AB, and the slope of any segment perpendicular to it. (This is the exception case — name it.)

(b) For $P(2, -4)$: determine the foot of the perpendicular from P to line AB, show that it lies *on* the segment, and state the shortest distance.

(c) For $Q(12, -1)$: determine the foot of the perpendicular, show it does **not** lie on the segment, and explain what that means for the shortest flight.

(d) Determine the shortest distance from Q to the segment, exactly and to the nearest tenth, and name the point of the segment it reaches.

(e) Calculate PA and PB. They are equal — yet neither is the answer to (b). Explain what that equality does tell you about P, in one sentence.

18. *(Multiple Choice)* A quadrilateral has four equal sides **and** equal diagonals. The most complete description of it is

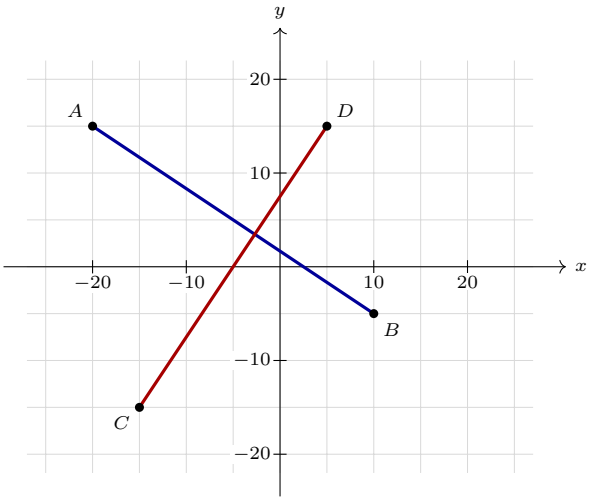
A. rhombus B. rectangle C. square D. not enough information

19. *(Numerical Response)* M is the midpoint of the segment joining $A(-5, 2)$ and $B(7, 10)$. The length of $\overline{AM}$, to the nearest whole unit, is _____. *(Record your answer in the numerical response box from left to right.)*

Part G — Cumulative Coverage

Every earlier set in this unit worked from coordinates that were handed to you, or from a grid whose squares were worth 1*. The booklet also teaches all three tools straight off a graph — including locating a midpoint by counting — and on a grid whose squares are not worth* 1 *the three tools do* ***not*** *all behave the same way. That is what this last question is about.*

20. On the grid below, **each square is** 5 **units across and** 5 **units up**. Read the tick labels, not the squares.



(a) State the coordinates of A, B, C and D.

(b) Determine the length of AB and of CD, exactly and to the nearest tenth.

(c) Determine the slope of AB and of CD, and say how the two segments are related.

(d) Locate the midpoint of AB and of CD **by counting on the grid**, then confirm each with the midpoint formula. Do the two segments share a midpoint? What does your answer rule out about $ACBD$?

(e) A classmate ignores the tick labels and treats every square as 1 unit. Which of your answers in (b), (c) and (d) would that student get *right*, and which wrong — and by what factor? Explain each verdict from the formula involved.

(f) Suppose instead the squares were 5 units across but only 1 unit up. Without recalculating anything, say which of length and slope would change, and which tool always locates the same grid point whatever the scale.